

ESTEBAN SCHAFIR

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ABOUT ME

I'm an applied AI engineer specializing in Large Language Models (LLMs) and a experienced cybersecurity professional. I have a proven track record of designing and building advanced AI solutions, from agentic Text-to-SQL engines to graph-based detection pipelines. I combine robust software engineering skills with hands-on experience in vulnerability management, penetration testing, threat detection, and security auditing.

SKILLS

- **AI & LLM Engineering:** Large Language Models (LLMs), Prompt Engineering, Retrieval-Augmented Generation (RAG), Agentic Workflows, In-Context Learning, Explainable AI (XAI).
- **Cybersecurity:** Vulnerability Assessment, Threat Modeling, Penetration Testing, Compliance Auditing (SIEM, ISO 27001), Security Risk Management, Identity & Access Management.
- **Languages & APIs:** Python, Java, C#, SQL, FastAPI, Flask, RESTful APIs, OpenAI/Anthropic/DeepSeek APIs.
- **Libraries & Frameworks:** PyTorch, TensorFlow, Hugging Face (Transformers, PEFT), LangChain, LlamaIndex, Scikit-learn, Pandas, NumPy.
- **Developer & Security Tools:** Docker, Git, GitHub, GitLab, Nmap, Nessus, Wireshark, Metasploit, SIEM (Splunk, SIEM tools), ITSM.
- **Database Management:** SQLite, MySQL, PostgreSQL, MongoDB, Vector Databases (Chroma, Pinecone).
- **Soft Skills:** Problem-solving, Teamwork, Technical Writing, Project Management.

RELEVANT EXPERIENCE

Applied AI & Security Research Assistant

Apr 2022 – Present

Florida International University, Miami, United States

- Developed a teacher-student distillation framework compiling online multi-step reasoning of LLM-based retrievers into offline visual sparse indexes for sub-10ms visual document search.
- Engineered and deployed end-to-end LLM pipelines, agentic RAG systems, and graph-based neural network models using PyTorch, Hugging Face, and LangChain.
- Designed a hybrid network alignment framework integrating Graph Autoencoders (GAEs) with seed-based percolation algorithms to optimize graph matching accuracy under structural perturbations.
- Architected machine learning solutions for CNN-based CAPTCHA bypass and multimodal bot detection to proactively identify and patch critical web vulnerabilities.

Information Security Officer

Sep 2017 – Nov 2021

Equifax Argentina SA, Buenos Aires, Argentina

- Directed enterprise-level security programs, managing comprehensive risk assessments, security control implementations, and third-party vendor security reviews.
- Spearheaded a company-wide security awareness initiative, boosting staff engagement and significantly reducing security risks.
- Slashed critical infrastructure vulnerabilities by over 50% by establishing and executing strategic remediation programs.
- Proactively identified and mitigated emerging security threats, strengthening the organization's global security posture and compliance alignment.

Information Security Analyst

Mar 2016 - Sep 2017

Puente Hnos, Buenos Aires, Argentina

- Orchestrated a rapid, secure enterprise-wide antivirus migration, earning recognition from executive leadership for execution excellence.
- Drove a 70% reduction in internal audit findings by streamlining mitigation workflows and serving as the primary security subject matter expert.
- Implemented and managed advanced security tooling (SIEM, ITSM, and privileged access management), reinforcing security infrastructure against unauthorized access.

Cybsec By Deloitte, Buenos Aires, Argentina

- Cut client security audit findings by 60% through targeted compliance reviews and system hardening across diverse client environments.
- Collaborated with corporate clients to resolve security policy violations and design customized risk mitigation strategies.
- Managed end-to-end vulnerability assessment lifecycles, leveraging Nessus, Nmap, Wireshark, and Metasploit to identify and patch system flaws.

EDUCATION**Ph.D in Computer Science**

2024 - Present

Florida International University, Miami, United States

GPA: 3.97/4.0

Masters in Cybersecurity

2023

Florida International University, Miami, United States

GPA: 4.0/4.0

BS in Programming

2017

Universidad Nacional de Quilmes, Buenos Aires, Argentina

PUBLICATIONS

- Chen, Z.; Salehi, H. A.; Schafir, E.; Zheng, X.; Zhang, J.; Wei, H.; Ni, J.; Shirani, F.; Luo, D. (2026) Addressing Structural Distribution Shift in Explanations for Graph Neural Networks. *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*, DOI: 10.1109/TPAMI.2026.3690304.
- Chen, Z.; Ni, J.; Salehi, H. A.; Zheng, X.; Schafir, E.; Shirani, F.; Luo, D. (2026) Explanation-Preserving Augmentation for Semi-Supervised Graph Representation Learning. *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*, DOI: 10.1609/aaai.v40i25.39183.
- Schafir, E.; Zheng, X.; Salehi, H. A.; Chen, Z.; Sha, M.; Cheng, W.; Luo, D. (2026) DecoSearch: Complexity-Aware Routing and Plan-Level Repair for Text-to-SQL. *arXiv preprint arXiv:2606.17821*.
- Zheng, X.; Chen, Z.; Schafir, E.; Chen, S.; Salehi, H. A.; Chen, H.; Shirani, F.; Sha, M.; Cheng, W.; Luo, D. (2025) Towards Structurally Explainable Machine-Generated Text Detection: A Graph-Perspective Framework. *arXiv preprint arXiv:2505.12507*.
- Tofighi, M. A.; Ousat, B.; Schafir, E.; Kharraz, A. (2024) Constructs of Deceit: Exploring Nuances in Modern Social Engineering Attacks. *Detection of Intrusions and Malware, and Vulnerability Assessment. DIMVA 2024*, DOI: 10.1007/9783031.6417186.
- Ousat, B.; Schafir, E.; Kharraz, A. (2024) The Matter of Captchas: An Analysis of a Brittle Security Feature on the Modern Web. *Proceedings of the ACM Web Conference 2024*, DOI: 10.1145/3589334.3645619.
- Jara, H.; Schafir, E. (2014) e-Health: An introduction to the challenges of privacy and security. *IEEE Central America and Panama Convention (CONCAPAN XXXIV)*, DOI: 10.1109/CONCAPAN.2014.7000407.

LANGUAGES

- **Spanish:** Native proficiency.
- **English:** Fluent (speaking, reading, writing).

PROFESSIONAL ACTIVITIES, PROJECTS & VOLUNTEERING

- **MightyPath (Drone Pathfinding App):** Developed an autonomous trajectory optimization application for drones, implementing real-time obstacle avoidance and path planning algorithms.
<https://github.com/eschafir/drone-mighty-path>
- **IEEE Membership:** Member since May 2024, # 98222076
- **Organizer, ISIT 2024 "Bits and Bots" Competition:** Served as a moderator for the competition at the IEEE International Symposium on Information Theory (ISIT) 2024.
<https://www.kaggle.com/competitions/isit-2024-bits-and-bots>
- **Photographer Volunteer, The GenAI Collective Miami**